
Literature Synthesis for the Alaoglu–Erdős Integer-Exponent Conjecture

A critical merge of four independent surveys, aligned with the remaining barriers in
`alaoglu-erdos-unified-report.tex`

Problem: $2^x, 3^x \in \mathbb{Z} \implies x \in \mathbb{Z}$

Inputs: four literature reports in `reports/`

Method: deduplication, source correction, barrier mapping, and ranked synthesis

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1 Executive synthesis

Bottom line. None of the four surveys contains an off-the-shelf theorem that proves the Alaoglu–Erdős conjecture, and none bypasses the same terminal obstruction isolated in the baseline report: a nontrivial integer-coefficient relation among products of logarithms of primes. The merge nevertheless changes the research portfolio in a useful way. It identifies two bodies of literature that are unusually close to the exact arithmetic geometry of a hypothetical counterexample, and three others that provide concrete mechanisms for creating the extra “source of size” missing from the present determinant arguments.

The five strongest clusters are as follows.

1. **Integer-valued entire functions on the four-exponentials lattice.** Pila’s two papers on entire functions sharing arguments of integrality, together with Pila–Rodríguez Villegas on concordant sequences, are the most direct literature found in any survey. They place exactly the four algebraic corner values $e^{a\alpha}, e^{a\beta}, e^{b\alpha}, e^{b\beta}$ on the semigroup $\{i\alpha + j\beta : i, j \geq 0\}$, and give an explicit quadratic-growth threshold below which a third integer-valued entire function must be algebraically dependent on e^{az} and e^{bz} . For the present problem the threshold is explicit. The immediate question is therefore numerical rather than philosophical: can the canonical interpolants already constructed in the baseline report be placed below that threshold, and if so can the resulting algebraic dependence be made contradictory? [provenance: R4, R3]
2. **Arithmetic holonomy and irrationality of products of logarithms.** Calegari–Dimitrov–Tang prove genuine irrationality and linear-independence results for certain products $\log(1 + 1/m) \log(1 + 1/n)$, using arithmetic holonomy and an integrated growth argument. Their theorem does not include $\log 2 \log 3$, and it must not be quoted as if it did. Its importance is methodological: it is a recent successful instance in which a product-of-logarithms problem acquires a second arithmetic size constraint, precisely the sort of phenomenon the baseline report lacks. [provenance: R3, R4]
3. **Multivariate, composite-ideal, and adelic generalized factorials.** The baseline report convincingly closes the plain one-dimensional, prime-by-prime Bhargava factorial route. It does not close the extensions found by the first survey: multivariate fixed divisors on the actual two-dimensional orbit grid; orderings for composite integers and arbitrary ideals; adelic regular bases; and matroid/greedoid structure of optimal ordering sets. These are the most testable new arithmetic leads because they can be implemented on finite orbit grids and assessed at the exact asymptotic scale required by the determinant deficit. [provenance: R1]
4. **Global and p -adic determinant architecture.** Heath-Brown, Salberger, Walsh, and recent global-field refinements explain how congruence coincidence can be converted into an auxiliary algebraic relation rather than merely summed term by term. Their theorems apply to algebraic varieties, not to the transcendental arc $y = x^{\log_2 3}$, so they are templates rather than plug-ins. They nevertheless give the right design discipline for Priority A: local divisibility should be used to force a global rank drop before the archimedean estimate is taken. [provenance: R3, R4]
5. **Conjugate control, house bounds, and adelic capacity.** Dimitrov’s Schinzel–Zassenhaus theorem, Pritsker’s variants, and the transfinite-diameter machinery of DeMarco–Rumely offer a possible route from the Newton–Kummer algebraic integer to a contradiction. They become relevant only after one controls enough conjugates or reciprocity; without that extra step they are comparison theorems, not solutions. The same capacity formalism may also prove a no-go theorem showing that a proposed determinant family is exactly capacity-critical. [provenance: R1, R4]

Recommended first moves

The merged evidence supports four immediate calculations: (i) compute the precise order-two growth constant of the canonical entire interpolant on Sol; (ii) compute multivariate and composite-ideal factorial defects on finite orbit grids and determine their leading scale; (iii) prototype a determinant whose 2- and 3-adic divisibility is extracted globally rather than termwise; and (iv) audit the Calegari–Dimitrov–Tang proof to identify exactly which part of its holonomy construction excludes the pair $(\log 2, \log 3)$. Each calculation has a clean stop condition and can therefore eliminate a direction without another large speculative program.

1.1 What the merge rules out

Several attractive-sounding routes in the supplied reports do not survive contact with the exact hypotheses.

- A hypothetical counterexample is necessarily irrational, since rational nonintegral exponents are ruled out by unique factorization. Results devoted only to rational x add no leverage.
- The set $\{(2^t, 3^t) : t \in \mathbb{R}\}$ is a transcendental analytic curve, not an algebraic curve in $\mathbb{G}_m^2$. Bombieri–Masser–Zannier, Maurin, Zilber–Pink, and arithmetic unlikely intersection theorems do not apply directly merely because the orbit points are S -integral.
- Classical Baker theory controls linear forms in logarithms with algebraic coefficients. It does not control the terminal coefficients $m_p \log 3 - a_p \log 2$, which are themselves logarithmic.
- The formal power series $(1 - z)^\theta$ with transcendental $\theta = \log_2 3$ is not a G -function in the standard André sense, because its coefficients are not algebraic. Arithmetic algebraization theorems for G -functions cannot be invoked without first constructing a genuinely arithmetic replacement.
- Generic rational-point counting tolerates far more points than the compact orbit produces. It must be strengthened to see prime support and synchronized denominator exponents; effectivity alone is not enough.

1.2 Classification used throughout

Every lead is marked by one of the following classes.

Mark	Class	Meaning in this report
D	Direct	The theorem is formulated in the four-exponentials lattice or in an essentially identical integer-valued setting.
A	Adaptation	The theorem could apply after one explicit, nontrivial lemma whose content is stated.
T	Template	The proof architecture is relevant, but the published hypotheses are materially different.
X	Diagnostic	The result identifies a frontier, obstruction, or reason not to spend effort on a route.
C	Computational	The literature supplies algorithms, finite certificates, or formalization infrastructure rather than an asymptotic theorem.

2 Baseline, notation, and the exact wall

This synthesis treats the attached unified report as the exclusion baseline. It therefore does not re-present the extensive structure theory, determinant calculations, Lean formalizations, or failed approaches already developed there. The minimum notation needed to evaluate the new literature is recorded here.

Set

$$\theta = \frac{\log 3}{\log 2}.$$

A counterexample is a positive irrational x for which both 2^x and 3^x are positive integers. Under the least-generator normalization of the baseline report, let β be the least positive nonintegral solution and put

$$M = 2^\beta, \quad A = 3^\beta.$$

The conditional solution and candidate monoids then have the rigid forms

$$\text{Sol} = \mathbb{N}_0 + \mathbb{N}_0\beta, \quad \text{Cand} = \{2^n M^k : n, k \geq 0\}.$$

The full rectangular orbit grid is

$$S_K = \{(2^a 3^b, M^a A^b) : 0 \leq a, b < K\} \subset \mathbb{Z}^2,$$

whereas the compact normalization is

$$U_k = \frac{M^k}{2^{\lfloor k\beta \rfloor}} = 2^{\{k\beta\}} \in \mathbb{Z}[1/2], \quad V_k = \frac{A^k}{3^{\lfloor k\beta \rfloor}} = 3^{\{k\beta\}} \in \mathbb{Z}[1/3].$$

Thus (U_k, V_k) lies on the compact analytic arc

$$\mathcal{A} = \{(2^t, 3^t) : 0 \leq t < 1\}.$$

The primitive output equation is

$$\log M \log 3 = \log A \log 2.$$

After prime factorization $M = \prod_p p^{m_p}$ and $A = \prod_p p^{a_p}$, it becomes

$$D = \sum_p (m_p \log 3 - a_p \log 2) \log p = 0. \tag{1}$$

This is an integer-coefficient linear form in products of two prime logarithms. It is not a linear form in logarithms with algebraic coefficients, and hence is not covered by Baker's theory. The literature is useful only if it does one of the following before reaching [Equation \(1\)](#):

- (a) creates an additional exact algebraic relation;
- (b) creates a second, independent lower bound or growth constraint;
- (c) converts synchronized local divisibility into a global determinant factor;
- (d) controls a sufficiently rich family of algebraic conjugates; or
- (e) exploits positivity, integrality, or prime support in a way unavailable for a general four-exponentials matrix.

2.1 Barrier model

The four surveys use different vocabularies. The following common barrier model is used in the synthesis and in the crosswalk tables.

Barrier	Exact content
B1: synchro-nized denomi-nators	A useful estimate must depend jointly on the 2- and 3-adic data, or on their common floor parameter, rather than separately optimizing two incompatible local problems.
B2: rank col-lapse	A counterexample produces a multiplicative semigroup of rank at most two. Existing rank theorems do not force rank one in the balanced 2×2 case.
B3: log-product wall	The final identity is Equation (1). A classical linear-form estimate loses algebraicity of the coefficients exactly here.
B4: p-adic amplification	A determinant needs local divisibility on the same leading scale as its archimedean size. Factoring rowwise minima or summing termwise tropical bounds is too weak.
B5: conjugate compatibility	Kummer conjugates of an algebraic radical do not transport the selected real logarithm relation equivariantly. Norms alone recover known identities.
B6: effective height gap	Available irrationality measures are polynomial in height, while the integral gap created by the candidate equations is exponentially small in the natural parameter.
B7: support-sensitive count-ing	The compact arc carries only logarithmically many ordinary rational points, so generic o-minimal bounds are vacuous. One must count points with disjoint prescribed prime supports and synchronized exponents.
B8: full-grid sparsity	Ordinary interpolation on a K^2 -point grid uses one monomial too many, or loses a comparable factor in degree. The desired relation must save support or exploit arithmetic not measured by support.
B9: local–global opti-mization	Separately optimal local orderings cannot simply be summed. The invariant must select rows, columns, and places jointly.

2.2 Crosswalk to the baseline research priorities

Priority	Baseline target	Most relevant new literature
A	p -adic amplification of semigroup determinants	Heath-Brown–Salberger–Walsh; multivariate/composite orderings; matroid defects; recent global determinant methods.
B	Full (n, k) -grid and sparse interpolation	Prasad–Rajkumar–Reddy; Evrard; multivariate fixed divisors; Pila’s geometric/arithmetic postulation.
C	Denominator-sensitive compact-arc counting	Campagna–Dill–Wilms; effective Pfaffian counting as a benchmark; analytic gcd bounds; composite ideal orderings.
D	Mixed archimedean–radical invariants	Dimitrov; Pritsker; DeMarco–Rumely; Bost slopes; adelic capacity.
E	Sharpen determinant hypotheses	Bombieri–Pila; global determinant method; Padé and potential theory; Hab-sieger’s alternants.
F	Force a third algebraic exponential	Pila’s third entire function; complex-power geometry only as a speculative existence framework.
G	Five/sharp six exponentials	Dasgupta’s rank survey; Roy’s parametric geometry; Waldschmidt–Masser frontier.
H	Larger effective/verified regions	Simons–de Weger; Padé/Hankel methods; DSLean-style certificate tooling.
I	Exact computational reconnaissance	Algorithms for generalized orderings; matroid intersection; finite determinant searches; digit/carry certificates.
J	Colossally abundant interface	Caveney–Nicolas–Sondow and the original Alaoglu–Erdős application.

3 Synthesis protocol and correction audit

3.1 How the four reports were merged

The reports were not concatenated. Their claims were normalized into a common record with five fields: theorem actually proved, hypotheses, proposed point of attachment to the baseline argument, missing lemma, and expected asymptotic scale.

Duplicate references were combined; incompatible assessments were resolved in favor of the narrower statement; and bibliographic metadata was corrected where the supplied reports disagreed with the cited source.

The provenance tags used below have the following meaning: R1 is the generalized-factorial and adelic-ordering survey; R2 is the broad transcendence/rank/dynamics survey; R3 is the products-of-logs and determinant-method dossier; and R4 is the integer-valued-entire-function, house-bound, and arithmetic-algebraization survey. Provenance records discovery, not endorsement.

3.2 Substantive corrections

Item	Claim appearing in a supplied report	Corrected assessment used here
Rational exponents	Results on rational x were presented as progress toward the conjecture.	Redundant. If $x = p/q$ in lowest terms and $2^x \in \mathbb{Z}$, unique factorization already forces $q = 1$.
Algebraic-curve methods	BMZ, Maurin, or Zilber–Pink were described as directly applicable to the orbit.	The orbit lies on a transcendental analytic arc. These theorems require an algebraic subvariety or an arithmetic family satisfying additional algebraic hypotheses.
Calegari–Dimitrov–Tang	Their product-of-logarithms theorem was suggested as nearly covering $\log 2 \log 3$.	It proves a nearby but different family involving $\log(1 + 1/m) \log(1 + 1/n)$, under a strong near-equality condition. It is a method lead, not a specialization.
G -functions	$(1 - z)^\theta$, $\theta = \log_2 3$, was called a G -function.	False in the standard arithmetic definition: the Taylor coefficients are generally transcendental. A separate arithmetic holonomic object would have to be constructed.
Dimitrov’s theorem	The house bound was stated without keeping track of reciprocal orientation.	Dimitrov’s theorem is naturally a statement about roots of an integer polynomial with $P(0) = 1$; using it for a Newton–Kummer algebraic integer requires a reciprocal polynomial or controlled inverse/conjugate relation.
Adelic regular bases	Applicability to the adelic closure of $\langle 2, 3 \rangle$ was stated as essentially automatic.	It is a concrete verification problem. The residual-cardinality and compactness hypotheses must be checked place by place, with 2 and 3 treated separately.
Composite 6-ordering	A 6-ordering was described as automatically coupling the two prime directions.	It couples them only through $v_{(6)}(n) = \min(v_2(n), v_3(n))$, which may discard rather than exploit an imbalance. It is a testable invariant, not a guaranteed gain.
Integer-valued o-minimal functions	arXiv:2404.10737 was attributed to Bhardwaj–van den Dries.	The paper is by Neer Bhardwaj, Raymond McCulloch, Nandagopal Ramachandran, and Katharine Woo.
Pila–Rodríguez Villegas	The paper was assigned to the wrong journal in one report.	<i>Concordant sequences and integral-valued entire functions</i> , <i>Acta Arith.</i> 88 (1999), 239–268.
Pila’s postulation paper	One report gave the wrong year.	<i>Geometric and arithmetic postulation of the exponential function</i> , <i>J. Austral. Math. Soc. Ser. A</i> 54 (1993), 111–127.
Pasten–Wang	The venue was listed as <i>Trans. Amer. Math. Soc.</i>	<i>GCD bounds for analytic functions</i> , <i>Int. Math. Res. Not.</i> 2017, no. 1, 47–95.
Complex powers	Exponential-algebraic closedness was described as a way to manufacture the missing independent column.	That is speculative and directionally opposite to the needed rigidity. The results prove existence/tameness, not arithmetic independence of a prescribed integral point.
Symbolic dynamics	Chaos or digit complexity was presented as evidence against a counterexample.	At most heuristic. Complexity becomes useful only after deriving an actual automatic, morphic, or finite-state sequence from the counterexample equations.

Interpretive rule

A paper can be highly relevant without its theorem applying. Throughout the report, “template” means that the proof contains a mechanism worth transplanting, not that the conclusion may be quoted for the Alaoglu–Erdős orbit. This distinction is especially important for determinant methods, unlikely intersections, G -function algebraization, and complex-power geometry.

4 Ranked research portfolio

The ranking below measures expected value for the particular barriers in [Section 2](#), not prestige or general mathematical importance. A high rank requires both structural proximity and a falsifiable next calculation.

Rank	Lead	Class	Barriers	Decisive next step
1	Pila, <i>Entire functions sharing arguments of integrality I–II</i> ; Pila–Rodríguez Villegas	D	B3, B8	Compute the order-two type of a canonical entire interpolant integer-valued on Sol ; compare it with $1/(832\beta^2 \log 2 \log 3)$; then classify the algebraic dependence forced below the threshold.
2	Prasad–Rajkumar–Reddy and Evrard: multivariate generalized factorials/-fixed divisors	A	B1, B4, B8, B9	Apply the sharp multidegree or total-degree factorial to the actual set $S_K \subset \mathbb{Z}^2$. Determine whether the mixed fixed divisor has a positive surplus on the leading determinant scale.
3	Calegari–Dimitrov–Tang: arithmetic holonomy and products of logarithms	T	B3, B6	Reconstruct the exact holonomic family, denominator estimates, and integrated-growth inequality; identify whether a deformation or logarithmic decomposition can include $(\log 2, \log 3)$ without destroying arithmeticity.
4	Lagarias–Yangjit: composite b -orderings and arbitrary-ideal orderings	A	B1, B4, B9	Compute orderings for (2), (3), (6) and a small family $(2^u 3^v)$ on candidate grids; compare the joint factorial with the sum of separate local optima at normalized leading order.
5	Grinberg–Petrov; Krachun–Mirgalimova: matroid/greedoid structure	A	B1, B9	Define the minimum simultaneous 2-/3-adic defect as a weighted matroid-intersection problem; prove or disprove a leading-scale lower bound.
6	Heath–Brown–Salberger–Walsh and global-field determinant refinements	T	B4, B8, B9	Replace termwise valuation bounds by a finite-point auxiliary hypersurface construction. The prototype succeeds only if local congruence forces a global rank drop before the transcendental arc enters.
7	Dimitrov; Pritsker: house bounds and capacity for algebraic integers	A	B5, B6	Produce a reciprocal or conjugate-stable Newton–Kummer polynomial and show that all relevant conjugates lie in the region where the house theorem contradicts the archimedean construction.
8	Chabert–Peruginelli: adelic orderings and regular bases	A	B1, B9	Verify the hypotheses for the adelic closure of $\langle 2, 3 \rangle$, compute the regular basis degrees, and test whether the global basis contains a mixed term absent from primewise factorials.
9	DeMarco–Rumely; Rumely: transfinite diameter and adelic capacity	T	B5, B9	Express the orbit determinant family as a capacity problem. Either obtain a strict product-formula inequality or prove that the family is capacity-critical and cannot close the missing logarithm.
10	Habsieger: alternants related to four exponentials	X	B3, B8	Translate the current semigroup determinant into Habsieger’s alternant language and isolate which positivity/divisibility condition would strengthen the known obstruction.
11	Bost slopes; Calegari–Dimitrov–Tang unbounded denominators	T	B3, B6, B9	Construct an arithmetic differential or holonomic object from the candidate lattice whose denominator growth violates a slope inequality. Without such an object, the theory has no input.
12	Schmidt–Summerer and Roy parametric geometry of numbers; shifted-log Padé systems	T	B3, B6	Encode the two simultaneous approximation problems as a combined successive-minima trajectory and locate an impossible slope pattern forced by exact integrality.
13	Simons–de Weger; explicit Padé/Hankel estimates; DSLean-style certificates	C	B6	Turn existing irrationality estimates for $\log 2/\log 3$ into compact kernel-replayable exclusion certificates over substantially larger finite ranges.
14	Adamczewski–Faverjon; function-field analogues; digit/carry literature	X	B2, B7	Derive an actual Mahler, automatic, or finite-state structure from the counterexample. If no such structure emerges, these remain analogies.
15	Caveney–Nicolas–Sondow and the colossally abundant interface	C	downstream	Formalize the implication from the conjecture to primality of successive colossally abundant quotients, after the core arithmetic modules stabilize.

Portfolio interpretation

Ranks 1–6 are active research programs with a finite first experiment. Ranks 7–12 require a new structural lemma before their literature can engage. Ranks 13–15 are valuable for verified finite progress, diagnostics, and the historical application, but they do not by themselves threaten the infinite conjecture.

5 Integer-valued entire functions on the solution lattice**5.1 Why this is the closest published program**

Suppose $a, b, \alpha, \gamma > 0$ and the four corner values

$$e^{a\alpha}, \quad e^{a\gamma}, \quad e^{b\alpha}, \quad e^{b\gamma}$$

are integers. Then the two exponentials

$$f_1(z) = e^{az}, \quad f_2(z) = e^{bz}$$

are integer-valued on the additive semigroup

$$X_{\alpha, \gamma} = \{i\alpha + j\gamma : i, j \in \mathbb{N}_0\}.$$

This is exactly the setup studied by Pila in *Entire functions sharing arguments of integrality I–II* [61, 62]. In the Alaoglu–Erdős normalization, take

$$a = \log 2, \quad b = \log 3, \quad \alpha = 1, \quad \gamma = \beta.$$

Then

$$e^{a\alpha} = 2, \quad e^{a\gamma} = M, \quad e^{b\alpha} = 3, \quad e^{b\gamma} = A,$$

and $X_{1, \beta} = \text{Sol}$. No change of category is required: the published theorem is already about the exact additive semigroup forced by a counterexample.

One of Pila’s order-two results implies, in this specialization, that an entire function g integer-valued on Sol and satisfying a sufficiently small quadratic-growth bound, for example

$$\limsup_{r \rightarrow \infty} \frac{\log M(g, r)}{r^2} \leq \frac{1}{832 \beta^2 \log 2 \log 3}, \quad (2)$$

under the corresponding hypotheses of the theorem, is algebraically dependent with 2^z and 3^z . Here $M(g, r) = \max_{|z| \leq r} |g(z)|$. The constant is not expected to be optimal; its value is that it turns an open-ended interpolation idea into a measured target.

Direct numerical test

For every canonical interpolant already present in the baseline program—Newton, q -Newton, Stieltjes, or a minimal-growth interpolation envelope—compute or rigorously bound

$$\tau_2(g) := \limsup_{r \rightarrow \infty} \frac{\log M(g, r)}{r^2}.$$

The first gate is whether $\tau_2(g)$ can be smaller than the right side of Equation (2). This calculation is meaningful even if it fails: the excess factor measures the same quantitative deficit that appears in the determinant hierarchy.

5.2 What algebraic dependence would and would not give

Pila’s conclusion is not by itself the Alaoglu–Erdős conjecture. It produces a nonzero polynomial

$$P(X, Y, Z) \in \mathbb{Z}[X, Y, Z]$$

such that

$$P(2^z, 3^z, g(z)) \equiv 0.$$

To obtain a contradiction, one needs one of the following stronger outcomes.

1. **Designed transcendence over the exponential field.** Construct g in a way that proves g is transcendental over $\mathbb{C}(2^z, 3^z)$. Then Pila’s dependence is impossible.
2. **Classification of low-growth algebraic branches.** Show that every entire function algebraic over $\mathbb{C}(2^z, 3^z)$ and having the relevant growth belongs to a small ring, ideally $\mathbb{C}[2^z, 3^z]$ or a finite algebraic extension with controlled ramification. Integer values on Sol, together with normalization conditions, may then exclude the resulting forms.
3. **A third algebraically independent interpolation condition.** Choose g to encode a quantity not recoverable from 2^z and 3^z , such as a carefully normalized finite-difference cocycle or a denominator-support statistic. This is the closest entire-function analogue of Priority F.

The second route deserves an explicit differential-algebra audit. Since 2^z and 3^z solve first-order linear differential equations, an algebraic relation involving an entire g may impose strong ramification and growth restrictions. Those restrictions are not supplied by Pila’s theorem, but they are potentially simpler than proving a new four-exponentials statement from scratch.

5.3 Concordant sequences and the minimal-growth envelope

Pila–Rodríguez Villegas construct minimal-growth transcendental entire functions integer-valued on broad classes of diffuse concordant sequences [60]. Geometric progressions are central examples, and the older Gel’fond–Fukasawa–Gramain theory gives sharp comparison scales for functions integer-valued on $\{a^n\}$ [38, 77]. There are two distinct node sets in the present problem:

$$\text{Sol} = \mathbb{N}_0 + \mathbb{N}_0\beta \quad \text{and} \quad \text{Cand} = \{2^n M^k : n, k \geq 0\}.$$

The first is additively two-generated but lies on the real line; the second is a multiplicative rank-two set whose logarithms are exactly $\text{Sol} \log 2$. The concordant sequence machinery should be tested on both representations, but its hypotheses must not be assumed. In particular, one should verify:

- whether an increasing enumeration of Cand is diffuse and concordant in the technical sense of the paper;
- whether the envelope construction preserves integrality on the full semigroup rather than only on the two coordinate progressions;
- how the envelope’s growth changes when the two generators have irrational logarithmic ratio; and
- whether the minimal envelope is visibly outside the algebraic closure of $\mathbb{C}(2^z, 3^z)$.

A promising computational approach is to construct finite Newton products over the first N nodes of Sol and Cand, normalize them by their fixed divisors, and estimate the logarithmic potential of the empirical node measure. If the limit exists, it should identify the true growth constant before a full concordance proof is attempted.

5.4 A concrete work package

Input.

The canonical interpolants and node-ordering machinery already developed in the baseline report; Pila’s exact hypotheses and constants; the concordant-sequence envelope.

Deliverables.

A rigorous upper and lower bound for $\tau_2(g)$; a proof or disproof of concordance for the chosen enumeration; and a classification lemma for entire functions algebraic over $\mathbb{C}(2^z, 3^z)$ in the attained growth class.

Success condition.

Either $\tau_2(g)$ lies below Pila’s threshold and the resulting algebraic dependence is excluded, or the comparison reveals a new structural constraint on any counterexample.

Stop condition.

If every natural integer-valued interpolant has a rigorously established type exceeding the threshold by a fixed factor and no algebraic-independence design is available, this route has quantified rather than removed the one-logarithm deficit and should be paused.

6 Arithmetic holonomy and products of logarithms

6.1 The exact result and its boundary

Calegari–Dimitrov–Tang prove new linear-independence statements involving products of two logarithms in their work on $1, \zeta(2)$, and $L(2, \chi_{-3})$ [16]. One consequence treats

$$\log\left(1 + \frac{1}{m}\right) \log\left(1 + \frac{1}{n}\right)$$

for nonexceptional integers m, n satisfying a stringent near-equality condition. In a representative form, the product is irrational, and when $m \neq n$ it participates with the two individual logarithms in a $\mathbb{Q}$ -linearly independent set.

The pair $(\log 2, \log 3)$ is not obtained by choosing m, n in this theorem. Indeed $\log 2 = \log(1 + 1)$ corresponds to an excluded boundary value, while $\log 3$ is not a single logarithm of the required neighboring form. Nor does a formal decomposition such as

$$\log 3 = \log 2 + \log \frac{3}{2}$$

solve the problem: it introduces cross-products and changes the arithmetic holonomic family. The theorem must therefore be treated as evidence that product-logarithm irrationality can sometimes be proved, not as a near-complete special case of the present conjecture.

6.2 Why the method matters

The relevant feature of the proof is the creation of two independent controls on the same auxiliary object:

1. a denominator or integrality estimate coming from arithmetic holonomy; and
2. an archimedean integrated-growth estimate strong enough to force a nonzero linear form to be too small.

This is conceptually parallel to the baseline determinant program, where integrality gives $|D| \geq 1$ but the analytic estimate misses by one logarithm. The CDT machinery succeeds in a family because holonomy supplies more structure than arbitrary logarithms possess. The research question is therefore not “can their theorem be substituted?” but “can the Alaoglu–Erdős lattice be encoded by an arithmetic holonomic family with comparable local and global controls?”

6.3 Four attachment experiments

6.3.1 Deformation through neighboring logarithms

Search for an identity expressing the terminal quantity as a specialization or limit of periods in the CDT family. A useful deformation must preserve all three properties:

arithmetic coefficients, uniform denominator control, uniform nondegenerate holonomy.

A mere analytic limit is insufficient, because irrationality and linear independence are not closed under specialization. The key output of this experiment is a proof that the necessary estimates remain uniform, or a precise statement of where they blow up as the parameters approach the desired boundary.

6.3.2 Telescoping decompositions

Since

$$\log q = \sum_j \log\left(1 + \frac{1}{m_j}\right)$$

can sometimes be achieved by telescoping products of neighboring rationals, one can ask whether both $\log 2$ and $\log 3$ admit decompositions using a common near-diagonal family. The obstacle is combinatorial explosion of cross-products. A successful decomposition would have to place all cross-terms inside a single holonomic period matrix, not simply invoke pairwise irrationality statements.

6.3.3 Holonomic encoding of the semigroup determinant

The determinant entries

$$(2^n M^k)^a (3^n A^k)^b$$

are hypergeometric in each integer index. One should test whether selected minors, their finite differences, or their generating series satisfy a Picard–Fuchs or holonomic system with arithmetic coefficients independent of the hypothetical transcendental parameter β . The last qualification is essential: a differential equation whose coefficients contain M , A , or β transcendently has not gained arithmeticity.

6.3.4 Unbounded denominators as a contradiction

The CDT proof of the unbounded denominators conjecture for modular forms [17] belongs to a broader slope/algebraization tradition. If a natural series attached to the candidate grid has bounded denominators because all its sampled values are integral, while arithmetic holonomy forces unbounded denominators unless the series is algebraic, then the candidate equations may force algebraicity. The missing object must be specified. The formal binomial series $(1 - z)^\theta$ is not suitable because its coefficients are not algebraic.

Stop condition for the holonomy route

Pause the program if every plausible generating series either has transcendental coefficients from the outset, loses uniform denominator bounds under specialization, or has a holonomy group too small to encode the product $\log 2 \log 3$. “Holonomic” without arithmetic coefficients is not progress on Equation (1).

6.4 Bost’s slope method and arithmetic algebraization

Bost’s algebraicity criteria for formal-analytic leaves and later arithmetic slope methods provide the general local–global language behind these ideas [11, 12]. An evaluation map on sections has an archimedean operator norm and nonarchimedean denominator bounds; the slope inequality constrains how often it can vanish without forcing algebraicity. For the present problem, an implementation would require:

- an algebraic ambient variety defined over a number field;
- a formal or analytic leaf whose Taylor coefficients are arithmetic;
- evaluation points or jets derived from Sol or S_K ; and
- a global height inequality in which the synchronized 2- and 3-adic gains are visible before separate optimization destroys them.

The baseline interpolants currently satisfy the analytic part of this template but not the arithmetic-leaf part. Establishing an arithmetic differential equation is therefore the first gate, not a technical afterthought.

7 Mixed generalized factorials, orderings, and fixed divisors

7.1 What the baseline report has already excluded

The baseline report develops ordinary Bhargava p -orderings for the scalar semigroup $\langle 2, 3 \rangle$, computes the separate local factorial behavior, and shows that the locally optimal orderings at 2 and 3 are incompatible. That work rules out the strategy

choose one scalar ordering $\longrightarrow$ sum independent p -adic factorial gains $\longrightarrow$ beat the archimedean determinant.

It does not rule out theories whose basic object is already multivariate, composite-ideal, or adelic. This distinction is the principal contribution of R1.

7.2 Multivariate fixed divisors on the actual orbit grid

Evrard introduced multivariate analogues of Bhargava factorials organized by total degree [30]. Prasad–Rajkumar–Reddy develop sharper fixed-divisor formulas for polynomials in several variables with multidegree and degree constraints [63]. Instead of replacing the counterexample by the one-dimensional set $\{2^a 3^b\}$, one can feed the theory the actual integral point set

$$S_K = \{(2^a 3^b, M^a A^b) : 0 \leq a, b < K\} \subset \mathbb{Z}^2.$$

Let $\Lambda \subset \mathbb{N}_0^2$ be a finite lower set of exponent pairs, and consider the evaluation matrix

$$E_{(a,b),(i,j)} = (2^a 3^b)^i (M^a A^b)^j, \quad (i, j) \in \Lambda.$$

The relevant generalized factorial controls the fixed divisor of every primitive polynomial

$$P(X, Y) = \sum_{(i,j) \in \Lambda} c_{ij} X^i Y^j$$

on S_K , or equivalently the unavoidable divisibility of suitable evaluation minors. This retains three pieces of information discarded by scalarization:

- the two coordinate functions have different prime supports at the compact level;
- the same pair (a, b) controls both coordinates;
- the Newton polytope Λ can be optimized against the triangular regions used by the baseline sparse-interpolation program.

The theorem itself does not promise a gain. The decisive statistic is a *mixed fixed-divisor surplus*. One useful operational definition is

$$G_{p,K}(\Lambda) := \min_T v_p \det E_T - L_{p,K}^{\text{flat}}(\Lambda), \quad (3)$$

where T ranges over full-rank choices of rows from S_K and $L_{p,K}^{\text{flat}}(\Lambda)$ is the best lower bound obtainable after replacing the row by a scalar magnitude and applying the already-analyzed one-dimensional ordering. The normalization can be adjusted to the determinant family, but the comparison must be made at the same cardinality and Newton polytope.

Success criterion 7.1. *The multivariate route is useful only if*

$$G_{2,K}(\Lambda_K) \log 2 + G_{3,K}(\Lambda_K) \log 3$$

has a positive contribution on the leading scale of the archimedean determinant deficit. An $O(K)$ boundary correction cannot repair a $\Theta(K^2)$ loss; in a determinant with $N \asymp K^2$ rows, the correct comparison may instead be N^2 , depending on the chosen monomial family. The scale must be fixed before numerical evidence is interpreted.

7.2.1 Finite experiment

For $K \leq 10$ and a collection of lower sets Λ —rectangles, total-degree triangles, and the baseline boundary-dilation shapes—compute:

1. exact Smith normal forms of the evaluation matrices;

2. the 2- and 3-adic valuations of all maximal minors or a certified minimum;
3. the multivariate factorial predicted by the fixed-divisor theorem; and
4. the normalized gap from the scalar primewise benchmark.

The Smith profile is more informative than one determinant: a systematic block of invariant factors would show that the gain survives row/column choices, whereas one exceptional minor may be a finite accident.

7.3 Composite b -orderings and arbitrary-ideal orderings

Lagarias–Yangjit extend Bhargava’s construction from prime ideals to composite integers and, more recently, to arbitrary ideals of Dedekind domains, including r -removed and order- h variants [49, 50]. For an integer $b > 1$, the basic divisibility order is

$$v_b(n) = \max\{e \geq 0 : b^e \mid n\}.$$

In particular,

$$v_6(n) = \min\{v_2(n), v_3(n)\}.$$

A 6-ordering therefore detects simultaneous divisibility of a difference by both primes. This is genuinely different from selecting a 2-ordering and a 3-ordering independently, but it can also be weaker: a large valuation at one prime is ignored when the other is small.

The right test family is not just $b = 6$. Use

$$\mathcal{B}_H = \{2^u 3^v : 0 \leq u, v \leq H, (u, v) \neq (0, 0)\}$$

or, in ideal notation, the collection $\{(2^u 3^v)\}$. Each slope u/v probes a different compromise between the two local directions. A weighted envelope over this family may retain information lost by the minimum in v_6 .

For a finite node set T , define a normalized joint-ordering score schematically by

$$J_H(T) = \sup_{\lambda_{u,v} \geq 0} \left\{ \sum_{u,v} \lambda_{u,v} F_{(2^u 3^v)}(T) : \sum_{u,v} \lambda_{u,v} (u \log 2 + v \log 3) = 1 \right\}, \quad (4)$$

where $F_b(T)$ is the logarithmic generalized factorial at the ideal $\mathfrak{b}$. The normalization prevents a trivial gain from repeatedly counting the same local valuation. Formula Equation (4) is a proposed diagnostic, not a theorem from the cited papers.

Research question 7.1. *Does the ideal envelope for the orbit grid exceed the convex envelope of the separate (2)- and (3)-factorials by a positive leading-order amount? Equivalently, is there a nonlinear synchronization phenomenon, or do arbitrary-ideal orderings merely repackage the same two local valuations?*

7.4 Matroid and greedoid formulations of simultaneous defect

Grinberg–Petrov show that finite optimality structures inspired by p -orderings have a matroidal description, while Krachun–Mirgalimova establish a strong greedoid structure for removed orderings [39, 47]. Johnson gives algorithms for r -removed and order- h variants [44]. These results suggest replacing the yes/no question “is there a simultaneous ordering?” by an optimization problem on common sets.

For a finite ambient set $T \subset \mathbb{Z}$ and $B \subset T$, set

$$c_p(B) = v_p \left(\prod_{\substack{x, y \in B \\ x < y}} (x - y) \right), \quad c_p^\star(k) = \min_{\substack{B \subset T \\ |B|=k}} c_p(B).$$

The simultaneous defect of a k -set is

$$\delta_k(B) = (c_2(B) - c_2^\star(k)) \log 2 + (c_3(B) - c_3^\star(k)) \log 3,$$

and the best compromise is

$$\delta_k^\star = \min_{|B|=k} \delta_k(B). \quad (5)$$

The baseline no-simultaneous-ordering result says that $\delta_k^\star$ is eventually nonzero under an ordering formulation. It does not quantify $\delta_k^\star$, and it does not exclude common optimal *sets* at selected cardinalities.

A matroid-intersection algorithm can compute Equation (5) exactly for finite truncations, while exchange inequalities may yield asymptotic lower bounds. There are two possible payoffs:

- If $\delta_k^\star$ is small, the result is negative but clarifying: local conflict cannot supply the missing determinant scale.
- If $\delta_k^\star$ grows on the determinant scale, every common node set pays a quantified synchronization penalty. Depending on the sign in the chosen determinant normalization, that penalty can either force extra divisibility or prove that a proposed row-selection strategy is impossible.

Care is needed with orientation: p -orderings minimize certain difference valuations, whereas determinant amplification seeks large valuations after structured row/column operations. The computational experiment must trace exactly how the factorial enters the denominator or determinant bound rather than infer the sign from terminology.

7.5 Adelic orderings and regular bases

Chabert–Peruginelli develop adelic versions of ordering and regular-basis theory for rings of integer-valued polynomials [20]. Conceptually this is attractive because one chooses the global basis before decomposing into local factors. Let E_p be the closure of the candidate set in $\mathbb{Z}_p$, and let $\underline{E} = \prod_p E_p$ with suitable restricted-product conditions. A successful application would provide a regular basis for $\text{Int}(\underline{E}, \widehat{\mathbb{Z}})$ whose leading coefficients encode all places simultaneously.

The applicability checklist is concrete:

1. Determine the closures of $\langle 2, 3 \rangle$ and $\langle 2, M \rangle$ in $\mathbb{Z}_p$, including exceptional behavior at 2 and 3.
2. Verify the finiteness/residual-cardinality hypotheses required for a global regular basis. It is not enough to say that the subgroup generated by 2, 3 mod p is usually large; the exact condition in the theorem must be proved uniformly in degree.
3. Compute the first basis denominators and compare them with the product of ordinary Bhargava factorials.
4. Determine whether the basis is sensitive to the synchronized exponent pair or only to the adelic closure, which may forget the orbit ordering completely.

If the closure already factors as an unrestricted product of local closures, no new coupling will appear. Conversely, a global degree obstruction or a nonfactorizable leading coefficient would be precisely the kind of B9 invariant sought by the baseline program.

7.6 Equidistribution and optimal-ordering asymptotics

Frączyk–Szumowicz and Byszewski–Frączyk–Szumowicz study simultaneous p -orderings, volume minimization, and optimal equidistribution in number fields [32, 14]. These results help answer a question that finite computations alone cannot: what limiting measure do near-optimal node sets approach? If the 2-optimal and 3-optimal measures are distinct, the minimum common defect may be approximated by an energy-transport cost between them. This creates a bridge to the capacity methods of Section 9.

The useful output would be a variational formula of the form

$$\lim_{k \rightarrow \infty} \frac{\delta_k^\star}{s(k)} = \inf_{\mu} (I_2(\mu) - I_2^\star + I_3(\mu) - I_3^\star),$$

for the correct scale $s(k)$. A strictly positive infimum would establish an asymptotic synchronization cost; a zero infimum would close this family of approaches.

8 From local divisibility to a global auxiliary relation

8.1 The classical determinant-method pattern

Bombieri–Pila begin with points on a short analytic arc and an evaluation determinant of monomials. Analytic smoothness makes the determinant small; integrality then forces it to vanish, placing the points on a low-degree algebraic curve [10, 59]. Heath-Brown replaces part of the real smallness by p -adic divisibility, and Salberger globalizes the construction; Walsh sharpens the resulting curve bounds in a way that removes a logarithmic loss in that algebraic setting [41, 69, 78].

The reusable pattern is

$$\text{many congruent points} \implies \text{large local determinant valuation} \implies \text{global auxiliary hypersurface}.$$

It is stronger than proving a valuation lower bound for each determinant term. The hypersurface is selected after local clustering and linear algebra, so cancellation is a feature rather than an uncontrolled remainder.

For a rational evaluation determinant $D \neq 0$, the product formula gives, after clearing known denominators,

$$\log |D|_\infty = \sum_p v_p(D) \log p.$$

Thus a sufficient local–global inequality is

$$\sum_{p \in \mathcal{P}} v_p(D) \log p > \log |D|_\infty + \log Q_D, \quad (6)$$

where Q_D records any cleared denominator not already included in the local valuations. Then $D = 0$. Priority A asks for precisely such an inequality with $\mathcal{P}$ containing at least 2 or 3, on a determinant with only $O(\log X)$ strategically selected rows.

8.2 Why the published theorems do not directly apply

The curve $\mathcal{A} = \{(2^t, 3^t) : 0 \leq t < 1\}$ is analytic and definable but not algebraic. The full grid S_K is an integral point set, but it is not given as the bounded-height rational points on an algebraic variety of fixed degree. The determinant-method literature uses the algebraic degree at two critical places:

1. to ensure that an auxiliary hypersurface not containing the original variety cuts the point set in controlled degree; and
2. to iterate the covering argument without the auxiliary equations becoming arbitrary.

For the Alaoglu–Erdős problem the desired conclusion is itself an auxiliary algebraic relation on a transcendental orbit. Consequently, the first vanishing determinant is not the end of the argument. One must either obtain a relation with fewer monomials than the baseline sparse-interpolation lower bound, or obtain enough compatible relations over several blocks to force an identity.

8.3 A semigroup-adapted global determinant prototype

Let

$$P_{n,k} = (2^n M^k, 3^n A^k)$$

and choose a finite exponent set $\Lambda \subset \mathbb{N}_0^2$. For rows $R = \{(n_i, k_i)\}_{i=1}^N$, set

$$D(R, \Lambda) = \det \left[(2^{n_i} M^{k_i})^a (3^{n_i} A^{k_i})^b \right]_{i, (a,b) \in \Lambda}.$$

The baseline tropical lower bound uses

$$v_p(D) \geq \min_{\sigma} \sum_i v_p(E_{i, \sigma(i)}),$$

which cannot see cancellation between minimizing permutations. The global determinant method suggests a different sequence of operations:

1. Partition exponent pairs (n, k) into residue cells on which the unit parts of $2^n M^k$ and $3^n A^k$ admit p -adic analytic expansions.

2. Within a cell, replace rows by multivariate finite differences in n and k . This exposes powers of p coming from congruence and derivative order rather than from a common monomial factor.
3. Choose the monomial lower set Λ to match the finite-difference filtration, as in a Hilbert-function determinant method.
4. Use the resulting local rank filtration to select one global minor with large total valuation; only then apply the archimedean bound.

At primes $p \nmid 6MA$, one can restrict n, k to residue classes modulo the torsion of $\mathbb{Z}_p^\times$ and write unit powers through the p -adic logarithm. At $p = 2, 3$, the nonunit valuations are explicit linear functions of (n, k) , while the residual unit parts may be handled after a deeper congruence restriction. This division into valuation and unit coordinates is standard in p -adic analytic parameterization and should be made explicit before determinant searches are run.

Success criterion 8.1. *A successful prototype must produce, for a family with $N \rightarrow \infty$,*

$$\sum_{p \in \{2,3\}} v_p(D) \log p - \log |D|_\infty \geq c s(N)$$

with $c > 0$ and $s(N)$ at the leading scale of both terms, after all denominators and monomial normalizations are included. A gain caused solely by a removable common row or column factor is not counted.

8.4 How a vanishing minor could become a proof

Suppose $D(R, \Lambda) = 0$. Then a nonzero polynomial supported on Λ vanishes on R . There are three possible ways to upgrade this finite relation.

1. **Boundary propagation.** Select overlapping triangular blocks so that the nullspaces agree on their intersections. The boundary-dilation equations in the baseline report are designed for this purpose.
2. **Shift covariance.** The transformations $(n, k) \mapsto (n+1, k)$ and $(n, k+1)$ multiply coordinates by fixed diagonal factors. A minimal-support nullvector that persists under both shifts may force a Laurent-polynomial identity on the entire semigroup.
3. **Fewnomial contradiction.** Choose $|\Lambda|$ below the number of distinct points in a way that violates the baseline monomial lower bound, using extra local divisibility rather than generic interpolation.

The second route can be formulated in module language. Let I_R be the ideal of polynomials vanishing on a block R . One seeks a low-support element in an intersection of shifted ideals

$$I_R \cap T_n^{-1} I_R \cap T_k^{-1} I_R,$$

where T_n, T_k are the two diagonal semigroup actions. This makes the passage from a local determinant to a global relation a finite compatibility problem rather than a vague compactness argument.

8.5 Walsh’s missing logarithm: analogy and limit

Walsh removes a logarithmic factor from a bounded-rational-point estimate on algebraic curves [78]. Since the baseline report also misses by one logarithm, the analogy is tempting. The mechanisms are not interchangeable. Walsh exploits the Hilbert function and degree of a fixed algebraic curve to choose a better auxiliary polynomial and count its intersections. In the present problem there is no fixed algebraic curve of bounded degree; producing one is the objective.

The transferable question is narrower: does the baseline semigroup determinant use a rectangular monomial family where a Hilbert-function-adapted staircase would save a boundary layer? This is a concrete optimization problem. It should be tested simultaneously with the multivariate fixed divisors of Section 7, because a staircase that is archimedean-optimal may be locally poor, and vice versa.

8.6 Habsieger’s alternants

Habsieger’s paper is directly concerned with alternants around the four-exponentials conjecture [40]. Its value here is not a new unconditional rank theorem but a mature normal form for the determinant obstruction. The current semigroup matrices should be translated into that language for two reasons:

- to check whether positivity, ordering, or total-positivity hypotheses proved in the baseline report imply a special alternant inequality not available in the general setting;
- to avoid rediscovering a determinant transformation whose failure is already encoded in Habsieger’s examples.

A useful deliverable is a lemma-by-lemma dictionary between Habsieger’s alternants and the baseline matrices, ending with the first hypothesis that cannot be verified. That point is likely to be a sharper formulation of the missing arithmetic input.

8.7 Recent global-field refinements

Recent work of Binyamini–Cluckers–Kato and related authors sharpens determinant bounds uniformly over global fields and improves dimension-growth estimates [8]. These results reinforce two design choices:

1. keep all places in one global height calculation rather than bolt a p -adic estimate onto an archimedean determinant at the end; and
2. choose the monomial/Hilbert-function budget before estimating local determinants.

They do not remove the transcendental-curve mismatch. Their role is architectural and quantitative once an algebraic auxiliary family has been manufactured.

9 Conjugate control, house bounds, and adelic capacity

9.1 The Newton–Kummer input

The baseline report constructs shifted higher Newton–Kummer cocycles $\Xi_{N,m}$ that are algebraic integers. In a distinguished real embedding, an adaptive choice $N = \ell$, $m = \ell - 1$ gives

$$\log |\Xi_\ell| = -c_0 \ell + o(\ell), \quad c_0 = 0.905325 \dots,$$

while genuine integer exponents make the cocycle exactly zero. This is unusually strong archimedean information: a nonzero algebraic integer is exponentially small in a natural parameter. The obstruction is that the ambient radical field has degree ℓ^ρ , $\rho \in \{3, 4\}$, and the other conjugates can be $\exp(O(\ell))$. The norm lower bound is therefore far too weak, and ordinary character projections erase the cross-base cancellation.

The literature in this section is relevant only if it reduces the effective conjugate complexity. No house theorem can contradict one small embedding while unconstrained conjugates are allowed to be enormous.

9.2 Dimitrov’s Schinzel–Zassenhaus theorem

Dimitrov proves a sharp universal lower bound of Schinzel–Zassenhaus type [29]. In a polynomial formulation, if $P \in \mathbb{Z}[z]$ has degree n , $P(0) = 1$, and is not cyclotomic, then a root lies in

$$|z| \leq 2^{-1/(4n)}.$$

Applied to the reciprocal minimal polynomial of a non-root-of-unity algebraic integer α , this yields

$$\overline{|\alpha|} \geq 2^{1/(4n)},$$

where the left side is the house.

For Ξ_ℓ , this theorem is not directly contradictory: the known conjugate upper bound is exponentially larger than $2^{1/(4\ell^\rho)}$. The theorem becomes useful only after one of the following reductions.

1. **Reciprocal normalization with uniform annular control.** Construct an algebraic integer η_ℓ from Ξ_ℓ whose conjugates all lie in $e^{o(1)} \leq |z| \leq e^{o(1)}$, while the distinguished conjugate remains $e^{-c\ell}$. Then a reciprocal/house theorem can force a contradiction.
2. **Low-degree quotient.** Show that Ξ_ℓ , or a normalized ratio of two shifted cocycles, lies in a subfield of degree $O(\ell)$ rather than ℓ^ρ . The linear exponential smallness could then compete with a degree-sensitive lower bound.
3. **Controlled factor of the conjugate polynomial.** Prove that the minimal polynomial has a noncyclotomic factor of low degree all of whose roots inherit the archimedean cancellation. Character projections do not provide this factor, so a nonlinear operation is required.

Pritsker’s work on algebraic integers with conjugate symmetry and related house bounds may be closer to a normalized cocycle if a unit-circle symmetry can be forced [64]. Again, symmetry without an upper bound is insufficient: reciprocal roots may simply place one conjugate far outside the unit circle.

9.3 A conjugate-control ledger

Every proposed modification of the cocycle should be evaluated by the following ledger.

Quantity	Required bound
Degree	Preferably $O(\ell)$, or at worst a scale for which the lower bound competes with $e^{-c_0 \ell}$.
Distinguished embedding	Preserve $\log \eta_\ell \leq -c\ell + o(\ell)$ with $c > 0$.
Other archimedean embeddings	Uniform $o(\ell)$, ideally $O(1)$, after normalization.
Nonarchimedean denominators	None for an algebraic integer; otherwise fully included in the height/product formula.
Cyclotomic/unit exceptions	Excluded by a structural argument, not by numerical evidence.
Dependence on M, A	Must retain both outputs; a projection that leaves only one axis has lost the essential cancellation.

This ledger prevents a common false gain: dividing by a large algebraic factor improves the distinguished embedding but introduces a denominator of the same global height.

9.4 Transfinite diameter and resultant identities

DeMarco–Rumely give transformation laws for transfinite diameter under polynomial maps, with the resultant recording the arithmetic distortion [26]. Rumely’s adelic capacity and Fekete–Szegő theorems with local rationality conditions provide a global framework for sets of possible conjugates [67]. These results can be used in two complementary ways.

9.4.1 Positive use: forbid a conjugate configuration

Associate to each place v a compact set E_v containing every conjugate of a normalized cocycle. At the distinguished real place, E_∞ contains a tiny disk; at other places, integrality may give unit disks or explicitly enlarged disks. If the global capacity

$$\gamma(\mathbb{E}) = \prod_v \gamma_v(E_v)$$

is strictly less than one, only finitely many algebraic numbers of the allowed type can have all conjugates in the adelic set. A family of growing degree would then be impossible unless it falls into a special algebraic locus.

The hard step is making the small real disk influence enough conjugates. One tiny component among ℓ^ρ archimedean embeddings contributes too little. A successful construction must cause a positive proportion of embeddings to lie in controlled small sets, or reduce the degree before capacity is applied.

9.4.2 Negative use: certify capacity criticality

The same calculation can show that an entire determinant/cocycle strategy is incapable of winning. If the optimal adelic sets have global capacity exactly one, the product formula is at equilibrium; no rearrangement of the same local data can produce a strict contradiction. This would turn the baseline’s observed one-logarithm deficit into a theorem about the capacity of the orbit configuration.

A recommended computation is therefore:

1. derive the limiting archimedean logarithmic energy of the normalized cocycle or Vandermonde node measure;
2. compute the 2- and 3-adic capacities of the corresponding closures;
3. include the resultant/discriminant factors exactly; and
4. test whether the global product is < 1 , $= 1$, or > 1 .

This is also a disciplined way to integrate the equilibrium-measure calculations already present in the baseline report with established potential theory.

9.5 Capacity and generalized-ordering limits

The generalized factorials of Section 7 and transfinite diameter are two finite and asymptotic faces of the same phenomenon: products of pairwise differences determine both Vandermonde valuations and logarithmic energy. If the optimal-ordering measures at 2 and 3 converge, one can ask whether a single global equilibrium measure minimizes

$$I_\infty(\mu) + I_2(\mu) + I_3(\mu)$$

subject to the semigroup support constraints. A strict gap between the minimum over all measures and the minimum over orbit-realizable measures would be a genuine local–global obstruction. No cited paper supplies this semigroup-constrained variational theorem, but the combined literature suggests exactly how to formulate it.

Stop condition for house/capacity methods

If every normalized cocycle retains degree ℓ^ρ with only one controlled small embedding, and the adelic capacity calculation is at least one, then Schinzel–Zassenhaus bounds cannot see the construction. Further improvements to the numerical house constant would not change that conclusion.

10 Parametric geometry, Padé systems, and potential theory

10.1 Parametric geometry as a language for the bilinear wall

Schmidt–Summerer parametric geometry of numbers and Roy’s realization/classification of n -systems encode Diophantine approximation through the successive minima of a one-parameter family of convex bodies [71, 65]. The modern variational formulation of Das–Fishman–Simmons–Urbański makes spectra and extremal trajectories systematic [22].

For the Alaoglu–Erdős problem, one can form a lattice or pair of lattices encoding simultaneously

$$m \log 2 - n \log 3 \quad \text{and} \quad a \log M - b \log A.$$

The exact relation $\log M \log 3 = \log A \log 2$ couples the slopes of these two approximation problems. A PGN reformulation is useful only if integrality forces an impossible pattern in the combined successive minima. Merely restating that each logarithmic ratio has good continued-fraction approximants does not add information.

A concrete model is to introduce a four-dimensional lattice generated by rows containing

$$(1, 0, \log 2, \log M), \quad (0, 1, \log 3, \log A),$$

with anisotropic weights chosen so that short vectors correspond to simultaneous small linear forms. The terminal identity imposes a rank-one condition on the real 2×2 logarithm matrix, while unique factorization and primitive-output constraints impose integral restrictions on admissible lattice vectors. The desired theorem would say that no self-similar n -system realizes all those restrictions.

What PGN does not provide automatically

Parametric geometry classifies possible approximation exponents for arbitrary real vectors. The present vector is special because its exponentials are integers. Unless that arithmetic condition is translated into a restriction on the minima trajectory, PGN is only a change of coordinates for the same open four-exponentials case.

10.2 Shifted logarithms and Padé constructions

Kawashima–Poëls and recent work on shifted logarithms develop explicit Padé systems and linear-independence measures for families of logarithms [46, 42]. Related S -unit Padé constructions aim to control denominators and linear forms simultaneously [68]. These methods can contribute in two restricted ways:

1. improve effective lower bounds for combinations of $\log 2$ and $\log 3$, thereby extending finite verified regions; or
2. supply an auxiliary table whose denominator structure can be synchronized with the candidate semigroup, creating a second integer rather than merely a better real approximation.

The first payoff is Priority H and is realistic. The second would be genuinely new. To test it, choose Padé approximants to a vector of functions containing

$$\log(1 - z), \quad \log(1 - 2z), \quad \log(1 - 3z)$$

or a binomial family, specialize at rational points whose denominators are powers of 2 and 3, and track the exact denominator ideal. The specialization must produce a relation in which the unknown M, A enter algebraically; otherwise it reduces to a known estimate for $\log 2/\log 3$.

10.3 Hankel determinants and irrationality criteria

Bugeaud–Han–Wen–Yao and related work connect Hankel determinants, Padé approximation, and irrationality of power-series values [13]. The baseline report already studies Hankel/Loewner and Markov–Padé defects. The literature can sharpen that program by separating three questions:

1. Is the moment sequence nondegenerate, so that infinitely many Hankel determinants are nonzero?
2. What are the exact denominator valuations of those determinants at 2 and 3?
3. Does potential theory give an asymptotic constant small enough to contradict integrality after the denominator is cleared?

Nonvanishing alone is not enough. A useful Hankel determinant must have an arithmetic normalization whose denominator cost is smaller than its archimedean decay. This is the same local–global ledger as Equation (6).

10.4 Classical Padé potential theory

The baseline report derives logarithmic energies for projected lattice triangles and for a Stieltjes-type interpolant. Gonchar–Rakhmanov equilibrium measures and the Stahl–Totik theory of rational approximation provide established tools for identifying exact n th-root asymptotics [37, 75]. Beukers’s Padé constructions for binomial functions offer explicit comparison families [6].

This literature can answer a crucial diagnostic question: is the residual positive constant in the current Padé calculation an artifact of the chosen nodes, or is it the equilibrium energy of the relevant compact set and therefore asymptotically optimal? The workflow is:

1. identify the compact set and external field determined by the node distribution;
2. solve or cite the equilibrium problem;
3. derive the asymptotic zero distribution of the Padé denominator;
4. compute the exact exponential rate of the remainder; and
5. compare it with the arithmetic denominator rate.

If the existing construction already attains the equilibrium rate and still loses, no reweighting of the same support can help. A successful improvement would then have to change the arithmetic object or add a second orthogonality condition, not merely tune coefficients.

11 Denominator-sensitive counting and unlikely intersections

11.1 The quantitative mismatch with ordinary counting

The compact orbit

$$(U_k, V_k) = (2^{\{k\beta\}}, 3^{\{k\beta\}})$$

contains $\asymp \log B$ points of multiplicative height at most B , because logarithmic height grows linearly with k . A Pila–Wilkie estimate of the form $O(B^\varepsilon)$ therefore does not constrain the orbit at all: $\log B \ll B^\varepsilon$. Effectivizing the constant in the same estimate does not change this comparison.

The arithmetic information omitted by ordinary height is unusually rigid. Up to reductions controlled by fixed valuations of M and A , the first coordinate has denominator supported only at 2, the second only at 3, and both denominator exponents are governed by the same floor $\lfloor k\beta \rfloor$. A theorem capable of proving the conjecture would resemble

$$\#\{t \in [0, 1) : 2^t \in \mathbb{Z}[1/2], 3^t \in \mathbb{Z}[1/3], h(2^t, 3^t) \leq H, \text{ synchronized denominators}\} = o(H), \quad (7)$$

when H is logarithmic height. The hypothetical orbit gives $\asymp H$ points. No survey located a theorem with this support-and-synchronization sensitivity.

11.2 Arithmetic unlikely intersections in powers of $\mathbb{G}_m$

Campagna–Dill–Wilms formulate and prove cases of arithmetic unlikely-intersection bounds in powers of the multiplicative group, including a complete dimension-one result and partial dimension-two results [18]. Their framework is closer to the desired arithmetic packaging than generic o-minimal counting because subgroup complexity and integral models enter the bound.

There are nevertheless two mismatches.

1. The point set in the present problem lies on a transcendental analytic arc, not an algebraic subvariety of $\mathbb{G}_m^2$.
2. The key restriction is not merely membership in a finite-rank subgroup. It is the coordinatewise prime support together with a common exponent parameter.

A viable adaptation would first algebraize a finite orbit block by a low-degree auxiliary relation obtained from [Section 8](#), then apply an arithmetic unlikely-intersection bound to that auxiliary curve and the relevant subgroup schemes. This is a two-stage program: the new paper may strengthen the second stage, but it does not provide the first.

11.3 Effective Pfaffian counting

Binyamini–Jones–Schmidt–Thomas give effective versions of Pila–Wilkie-type counting for Pfaffian structures [9]. The arc $t \mapsto (2^t, 3^t)$ is well within such tame analytic settings. The result is useful for explicit bookkeeping once one has a counting exponent below the orbit density. By itself, it retains only height and definable complexity, so the bound remains too large.

The most plausible way to combine it with arithmetic is to partition rational points by exact denominator ideals and apply determinant or congruence arguments inside each fiber. However, there are only $O(H)$ relevant synchronized fibers up to logarithmic height H , and one orbit point per fiber is already enough. A successful theorem must show that most fibers are empty for arithmetic reasons; a uniform bound per fiber is not sufficient.

11.4 Integer-valued definable functions

Jones–Thomas–Wilkie and later work on integer-valued o-minimal functions establish strong polynomiality or growth dichotomies for definable functions taking integer values on the integers [45, 7]. This literature offers a clean conditional route:

Construct a function F definable in the required o-minimal structure, with controlled growth, such that $F(n) \in \mathbb{Z}$ for every $n \in \mathbb{N}$ and such that F encodes $n \mapsto n^\theta$ or the candidate semigroup. The definable integer-valued theorem may then force F to be a polynomial, which is incompatible with irrational θ .

The current canonical interpolants are selected through discrete minimization or infinite products on sparse nodes and are not known to be definable in a useful uniform structure. Thus the missing lemma is exactly the construction of a definable

representative. The literature should not be applied to x^θ merely because that function itself is definable: it is not integer-valued on every integer, only on the hypothetical candidate submonoid.

A more promising definable object may be an envelope or extremal function whose parameter set is continuous and whose integrality on Cand forces integrality on a definable family. This would convert sparse interpolation into a unary integer-valued problem, but no such extension principle is currently known.

11.5 Analytic gcd bounds

Pasten–Wang extend gcd estimates to analytic/meromorphic functions using Nevanlinna theory [57]. The baseline program contains gcd/lcm lattices, common-divisor heuristics, and pairs of interpolating functions. The theorem could become relevant if one constructs algebraically independent meromorphic functions F, G for which the counterexample forces large common divisors or many common zeros at the candidate nodes.

The required translation should be written before the theorem is invoked:

1. identify F and G and prove the algebraic-independence hypothesis;
2. express the integer gcd or common-zero multiplicity in the Nevanlinna counting function used by the theorem;
3. show that the candidate orbit forces a lower bound exceeding the theorem’s upper bound; and
4. include poles and denominator contributions exactly.

Finite differences of one interpolant are not automatically algebraically independent from it, and sharing a discrete set of integer values is not the same as sharing zeros. This is a conditional attachment point, not a direct gcd theorem for $M^n - 1, A^n - 1$.

11.6 Why generic BMZ, Maurin, and Zilber–Pink remain out of scope

Bombieri–Masser–Zannier and Maurin prove finiteness or structural results for intersections of an *algebraic* curve with algebraic subgroups of $\mathbb{G}_m^n$ [5, 54]. The map $t \mapsto (2^t, 3^t)$ does not define such a curve. Embedding its graph in a larger exponential-algebraic structure does not restore the algebraic degree needed by the theorem. These results become relevant only after an auxiliary algebraic curve has been forced, in which case they may help show that the curve cannot contain too many orbit points unless it is a subgroup translate.

Stop condition for counting

Do not pursue a counting theorem whose bound depends only on ordinary height and definable complexity. To beat the hypothetical orbit it must use denominator support, synchronized exponents, subgroup complexity after algebraization, or another arithmetic invariant that makes the right side of Equation (7) sublinear in logarithmic height.

12 Complex powers and model-theoretic existence results

Gallinaro proves exponential-algebraic closedness results for systems of raising-to-powers type, and Gallinaro–Kirby establish quasiminimality phenomena for structures with complex power maps; Aslanyan–Kirby–Mantova give related geometric existence theorems for exponential equations [33, 34, 4]. These papers are relevant because the distinguished power map

$$z \mapsto z^\theta, \quad \theta = \frac{\log 3}{\log 2},$$

is exponentially algebraic even though θ is transcendental.

Their conclusions point in the opposite direction from the desired theorem. They show that suitable systems have solutions and that the ambient structure is tame; they do not show that a prescribed solution with integer coordinates is impossible. In particular, an existence theorem for auxiliary powers does not manufacture a multiplicatively independent integer base q with controlled q^x .

There are still two diagnostic uses.

1. Formulate the counterexample locus in the language of complex powers and determine its predimension. If the integer-support constraints force an atypical predimension drop beyond what the existence theory permits generically, this may identify the exact additional arithmetic statement required.

2. Use quasiminimality/tameness to rule out proposed definability arguments that would make the set of candidates too complicated. This can prevent a false model-theoretic route, but is unlikely to yield a quantitative contradiction.

The appropriate classification is therefore **T/X**, not a proposed source of the missing independent column.

13 Function-field, Mahler, and digit/carry analogues

13.1 Positive-characteristic logarithm independence

Papanikolas’s Tannakian theory for Anderson–Drinfeld motives identifies transcendence degree with the dimension of a difference-Galois group, and Chang–Papanikolas determine algebraic relations among periods and logarithms of Drinfeld modules [56, 21]. Recent work of Gezmiş–Namoiyam extends parts of this picture to logarithms of Anderson t -modules [36]. In positive characteristic these are unconditional analogues of transcendence statements that remain open over $\mathbb{Q}$.

The mechanism replacing the missing archimedean size is not a stronger determinant estimate. It is Frobenius-semilinear difference structure: a rigid analytic trivialization satisfies a σ -difference equation with algebraic coefficients, and the dimension of the associated Galois group controls algebraic independence. This observation suggests a structural design criterion for characteristic zero:

A successful auxiliary object should possess a semilinear or differential symmetry whose Galois group remembers both logarithmic directions and whose arithmeticity is independent of the hypothetical transcendental parameter.

The maps $n \mapsto 2n$ and $n \mapsto 3n$, or the two shifts on the exponent grid, are obvious candidates for a difference structure. They do not behave like Frobenius: there is no coefficient-field endomorphism making 2^z and 3^z solutions of one common semilinear equation. A transfer from the function-field proof would therefore require a new characteristic-zero difference module, not an analogy at the level of statements.

A useful negative deliverable is to prove that the natural two-shift module generated by the candidate table has difference-Galois group of the expected rank unless $\beta \in \mathbb{Q}$. Such a theorem might repackage the conjecture but would at least reveal whether the missing condition is a Galois-group dimension statement.

13.2 Mahler’s method and automatic structure

Adamczewski–Faverjon give powerful theorems on linear and algebraic relations among values of Mahler functions at algebraic points, with effective variants [1, 2]. Adamczewski explicitly discusses the Alaoglu–Erdős problem in the surrounding Mahler/Cobham circle [3].

This literature becomes actionable only after deriving a functional equation such as

$$\mathbf{F}(z) = A(z)\mathbf{F}(z^q)$$

with $A(z)$ algebraic/rational and an algebraic specialization. The natural candidate sequences—digits of $[k\beta]$, carries in the 2- and 3-adic normalizations, or coefficients of an interpolant—do not automatically satisfy a Mahler equation. Irrational rotations are Sturmian rather than automatic in general.

There are three precise tests:

1. Does simultaneous integrality force the carry word to be 2-automatic or 3-automatic?
2. Does the same sequence admit compatible descriptions in both bases, so that a Cobham- type theorem forces eventual periodicity?
3. Can a generating function of the orbit be shown to be Mahler without coefficients containing M , A , or β transcendently?

If the answer to the first two questions is yes, eventual periodicity would likely imply $\beta \in \mathbb{Q}$, finishing the problem. If the sequence remains merely Sturmian, the Mahler literature is not engaged.

13.3 Ternary digits of powers of two and carry certificates

Lagarias studies ternary expansions of powers of two through 3-adic dynamics [48]. Saye gives recursive certificates for very large finite ranges of prescribed trailing ternary patterns, and Dimitrov–Howe analyze powers of 3 with few nonzero binary digits [70, 28]. These works do not imply the Alaoglu–Erdős conjecture, but they offer two transferable tools.

First, they show how multiplicative dynamics can be encoded as compact trailing-digit recursions rather than enormous integer computations. This is attractive for Priority I: a candidate search or zero-free verification can be accompanied by a short recursive certificate checked in Lean.

Second, the 3-adic Cantor-set perspective suggests studying the set of possible unit parts of M^k simultaneously in $\mathbb{Z}_2$ and $\mathbb{Z}_3$. The desired statement would be an intersection-dimension or entropy bound for an orbit constrained by two incompatible digit alphabets. Existing digit-omission theorems treat one base action at a time; they do not couple the common floor parameter.

The experimental target should therefore be a *synchronized carry automaton*. For a finite depth r , record the state

$$(M^k \bmod 2^r, A^k \bmod 3^r, \lfloor k\beta \rfloor \bmod q)$$

with transitions in k . Search for forbidden state combinations stable as r grows. Any discovered invariant must be converted into an exact congruence theorem; entropy or apparent chaos is not evidence by itself.

13.4 Recent gap/word results

Hultberg’s recent gap principles on semiabelian varieties and Stephan’s work on steering words are remote but potentially useful diagnostics [43, 76]. The former would require an algebraic semiabelian orbit of the correct type; the latter would require a finite-word encoding whose low complexity follows from the counterexample. Neither input is currently available. They belong in the lead inventory so that a future structural breakthrough can activate them, but they should not compete with the direct programs in Sections 5, 7 and 8.

Template lesson

The positive-characteristic and Mahler successes all possess a rigid functional equation over an arithmetic coefficient field. The Alaoglu–Erdős table has many shift identities, but their coefficients involve the unknown integral outputs. The decisive transfer problem is to eliminate that dependence while retaining both logarithmic directions.

14 Rank theorems and the four-exponentials frontier

14.1 Conditional resolution and unconditional boundary

The Four Exponentials Conjecture says that if x_1, x_2 and y_1, y_2 are each $\mathbb{Q}$ -linearly independent, then at least one of $e^{x_i y_j}$ is transcendental. Apply it to

$$(x_1, x_2) = (1, x), \quad (y_1, y_2) = (\log 2, \log 3).$$

If all four exponentials $2, 3, 2^x, 3^x$ are algebraic, Four Exponentials forces $x \in \mathbb{Q}$, and unique factorization then gives $x \in \mathbb{Z}$. Thus the conjecture is an exact special case of the four-exponentials frontier, not merely adjacent to it.

Dasgupta’s Part I gives a clean modern account of unconditional rank theorems for matrices of logarithms of algebraic numbers, including Baker and Waldschmidt–Masser results and Roy’s structural-rank perspective [25]. Dasgupta–Kakde’s Matrix Coefficient Conjecture makes the balanced 2×2 obstruction explicit [24]. These sources are important for calibrating claims: a proposed rank argument should be checked against the precise hypotheses and known lower bounds before being presented as new progress.

14.2 Quadratic independence of prime logarithms

Equation (1) is excluded by a suitable $\mathbb{Q}$ -linear independence statement for products $\log p \log q$. For example, linear independence of the relevant part of the symmetric square of the $\mathbb{Q}$ -space spanned by prime logarithms would force all coefficients m_p, a_p to vanish in the prohibited combination. This is an exact sufficient condition, but there is no evidence that it is easier than the original problem; it is a broad fragment of the four-exponentials circle.

The useful research question is narrower: do positivity and disjoint support of the exponent vectors in Equation (1) define a cone on which independence can be proved without settling arbitrary signed relations? None of the four surveys found such a theorem. This “positive cone” formulation should be kept visible because determinant, capacity, or convexity arguments may exploit it even when general quadratic independence remains open.

14.3 Analytic subgroup and period-matrix packaging

Wüstholz’s analytic subgroup theorem and quantitative period theorems of Gaudron–Rémond provide the natural algebraic-group language for unexpected rank drops [79, 35]. They are extraordinarily powerful when the logarithms or periods lie in the Lie algebra of an algebraic group and the relevant coefficients are algebraic. In the present matrix, the unknown logarithms $\log M$, $\log A$ are logarithms of algebraic numbers, but the rank-one relation involves the transcendental coefficient ratio $\log 3/\log 2$. The analytic subgroup theorem does not automatically lower the rank from two to one.

This packaging may nevertheless interface better with valuation lattices and 1-motives than raw linear forms do. A concrete adaptation would construct an algebraic subgroup from the integral semigroup action, not assume one from the real logarithmic relation. Until that subgroup is exhibited, the AST is a frontier theorem rather than a proof route.

14.4 Roy, Diaz, and the ceiling of current independence results

Roy’s small-value and parametric-geometry results sharpen the six-exponentials frontier, while Diaz proves substantial algebraic independence for iterated powers with algebraic exponents [66, 27]. These theorems show how much independence can be extracted when an algebraic exponent or a third direction is available. A hypothetical counterexample is balanced to remove exactly those extra degrees of freedom.

The correct use of these results is to formulate gateways. For example:

- any construction of a third multiplicatively independent integer q with algebraic q^x invokes six exponentials;
- any construction of a nonzero algebraic γ with algebraic $e^{\gamma x}$ invokes five-exponentials-type restrictions; and
- any reduction of β to an algebraic irrational invokes stronger independence theorems and likely gives an immediate contradiction.

Without such a gateway, quoting the theorem only restates why the two-base case is hard.

14.5 Triage of the broad rank/dynamics survey

The useful content of R2 is its clean conditional Four Exponentials reduction, its emphasis on structural rank, and its pointer to the matrix-coefficient framework. The following claims are downgraded:

- Baker-type improvements do not cross the logarithmic-coefficient wall.
- BCZ/S-unit finiteness does not imply nonexistence and usually assumes a fixed algebraic curve or recurrence.
- Symbolic chaos of fractional parts is heuristic unless exact automatic structure is derived.
- Generic unlikely intersections and the analytic subgroup theorem require algebraic input absent from the transcendental arc.
- Metric Bézout or small-value theorems need a new auxiliary construction; they do not geometrize the problem for free.

This correction is not a dismissal of the surrounding theories. It locates the precise new lemma each would need before it can touch the conjecture.

15 Effective finite progress, formal certificates, and the CA interface

15.1 Explicit approximation to $\log 2/\log 3$

Simons and de Weger develop effective continued-fraction and linear-form estimates involving $\log 2$ and $\log 3$ [73, 72]. These estimates do not address the bilinear relation Equation (1), but they can make finite exclusion substantially cheaper. The baseline verification tests, for each integer m in a range, whether

$$m^\theta, \quad \theta = \frac{\log 3}{\log 2},$$

can be an integer, using rigorous enclosures obtained from convergents and semiconvergents. A uniform irrationality/linear-form bound can replace many ad hoc precision tiers by one certificate schema.

The correct benchmark is not merely a larger floating-point scan. A useful result should produce a compact proof object of the form

$$\forall m \leq X, \quad \text{dist}(m^\theta, \mathbb{Z}) > \varepsilon_m > 0,$$

where every ε_m follows from a small list of rational interval enclosures and integer inequalities. The certificate must replay in the kernel without trusting MPFR rounding or an external search result.

Padé and hypergeometric improvements surveyed by Zudilin and developed by Marcovecchio may reduce the size of these certificates [80, 51]. Their role remains finite unless the resulting measure is strong enough to compare with the exponentially small gap generated by an arbitrary unknown M , which ordinary linear-logarithm measures are not.

15.2 Certificate translation and DSLean

DSLean provides infrastructure for translating certificates from external domain-specific solvers, including interval arithmetic/Gappa-style proofs, into Lean terms checked by the kernel [23]. The baseline project already has generated tables and semiconvergent enclosures. The natural integration is:

1. an external generator selects convergents, semiconvergents, and interval splits;
2. Gappa or an exact rational solver verifies each transcendental enclosure from a small library of analytic lemmas;
3. DSLean translates the certificate into a Lean proof term; and
4. one generic theorem lifts the per-row inequalities to the finite zero-free region.

This architecture separates trusted mathematics from untrusted search while avoiding a monolithic decide table. It is also reusable for determinant sign checks, capacity constants, and Padé remainder inequalities.

15.3 Digit recursion as proof compression

Saye’s trailing-ternary-digit recursion verifies enormous finite ranges using a short recursive construction rather than enumerating every exponent [70]. A comparable compression may be possible for the Alaoglu–Erdős scan if candidate intervals can be organized by a common prefix in the binary expansion of $\theta \log_2 m$ or by a common semiconvergent tier. The output should be a tree whose internal node carries one interval inequality proving all descendant values impossible. This is more promising than simply raising the numerical frontier because the same tree can be replayed and minimized.

15.4 The colossally abundant consequence

The historical motivation should be kept logically separate from the proof search. Work of Caveney–Nicolas–Sondow records the consequence that, under the Alaoglu–Erdős assumption, quotients of consecutive colossally abundant numbers are prime [19]. The original Alaoglu–Erdős analysis already restricts such quotients sharply.

This is an excellent formalization target because it is downstream and combinatorial. A clean module should state:

1. the definition and ordering of colossally abundant numbers;
2. the formula determining their prime exponents from a parameter;
3. the reduction of a composite consecutive quotient to a rational-power equality; and

4. the invocation of the integer-exponent conjecture to force primality.

Formalizing this implication will not advance the transcendence barrier, but it gives a machine-checked historical payoff and tests whether the conjecture has been stated in the strongest form needed by the application.

Role of computation

Finite verification should be optimized for theorem discovery and proof compression. The highest-value computations are those that estimate a leading asymptotic constant, expose a Smith-profile pattern, or produce a reusable exact certificate. Extending a decimal frontier without changing the structure remains evidence only.

16 Integrated research program

The merged literature supports a staged program in which every work package has a binary or quantitative decision gate. The packages are ordered to maximize information gained before large formal or theoretical investments.

16.1 WP1: Pila growth audit

Goal. Decide whether the canonical entire-function program enters Pila’s direct order-two dependence theorem.

Tasks.

1. Select two canonical functions: one interpolating on Sol, one on Cand.
2. Prove maximum-modulus upper bounds using the exact node-counting measure, not a generic Hadamard product estimate.
3. Obtain a matching lower bound from the concordant envelope or logarithmic potential.
4. Compare the resulting type with Equation (2).
5. If below threshold, classify the algebraic dependence over $\mathbb{C}(2^z, 3^z)$.

Decision gate. Continue only if the threshold is met, approached within a structurally improvable factor, or the algebraic-dependence classification yields a new constraint. Otherwise record the exact excess constant as a no-go benchmark.

16.2 WP2: mixed-factorial scaling experiment

Goal. Determine whether multivariate/composite/adelic orderings create a leading- scale synchronization term absent from scalar primewise factorials.

Tasks.

1. Implement the Prasad–Rajkumar–Reddy and Evrard factorials for lower sets $\Lambda \subset \mathbb{N}_0^2$.
2. Compute exact Smith forms for S_K with symbolic valuation parameters where possible and concrete primitive pairs otherwise.
3. Implement b - and ideal-orderings for $b = 2^u 3^v$.
4. Solve the simultaneous-set defect Equation (5) using matroid/greedoid algorithms.
5. Fit normalized asymptotics and prove the first stable recurrence or energy formula.

Decision gate. Continue only if a mixed gain survives after normalization at the archimedean determinant scale. A boundary-order gain is insufficient.

16.3 WP3: global p -adic determinant prototype

Goal. Turn semigroup congruence into an auxiliary polynomial before applying the archimedean bound.

Tasks.

1. Develop p -adic analytic coordinates for exponent cells at $p = 2, 3$ and at one generic odd prime.

2. Design finite-difference row bases matched to staircase monomial sets.
3. Prove a local rank-filtration lemma analogous to the p-adic determinant method.
4. Combine places globally and calculate every denominator in Equation (6).
5. Test shift covariance of the resulting nullvectors on overlapping blocks.

Decision gate. The first prototype must beat the tropical termwise ceiling on the leading scale. If a general inequality proves that it never can, promote that inequality to a no-go theorem and stop this family.

16.4 WP4: CDT holonomy suitability audit

Goal. Determine whether arithmetic holonomy can be made to see the $(\log 2, \log 3)$ product rather than a neighboring family.

Tasks.

1. Reproduce the product-of-logarithms theorem with all constants and identify the source of the near-diagonal restriction.
2. Test telescoping logarithmic decompositions and parameter deformations for uniformity.
3. Search for an arithmetic holonomic generating series of semigroup minors or Padé defects.
4. Verify coefficient algebraicity and denominator growth before any analytic estimate.

Decision gate. Continue only if the desired specialization remains inside a uniform arithmetic family. Analytic approximation by admissible parameters is not enough.

16.5 WP5: Newton–Kummer conjugate compression

Goal. Reduce the effective degree or control a positive proportion of conjugates of the exponentially small cocycle.

Tasks.

1. Search nonlinear resultants, ratios of adjacent shifts, and norm-one normalizations that retain both M and A .
2. Track the conjugate-control ledger of Section 9 exactly.
3. Compute local and archimedean capacities of the normalized conjugate sets.
4. Apply Dimitrov/Pritsker only after the degree and annular bounds compete with $e^{-c_0 \ell}$.

Decision gate. Stop if only one of ℓ^p embeddings is small and global capacity remains at least one.

16.6 WP6: certified finite expansion and historical interface

Goal. Extend the kernel-replayed region while improving proof architecture.

Tasks.

1. Integrate explicit continued-fraction bounds with a DSLean/Gappa certificate translator.
2. Replace flat tables by recursive interval or digit certificates.
3. Use the same infrastructure for numerical constants in WPs 1–5.
4. Formalize the colossally abundant quotient implication in a separate downstream module.

Decision gate. A new finite range is worth merging only if its certificate size and replay cost scale substantially better than row-by-row enumeration.

16.7 Dependency graph

Package	Can start immediately?	Dependencies / shared outputs
WP1 Pila	Yes	Shares potential-theory constants with WP5 and exact interpolation code with WP2.
WP2 factorials	Yes	Supplies local divisibility data to WP3 and equilibrium measures to WP5.
WP3 determinant	Yes, finite prototype	Benefits from WP2’s optimal lower sets and WP6’s certificate engine.
WP4 holonomy	Yes, source audit	May supply a global algebraization mechanism to WP1 or WP3.
WP5 conjugates	Partly	Needs normalized cocycles from the baseline; capacity calculation can use WP2 measures.
WP6 certificates	Yes	Independent infrastructure; should not consume core research effort beyond reusable tooling.

16.8 Allocation recommendation

The first research cycle should divide effort approximately as follows: one third on WP1, one third split between WP2 and WP3, one sixth on the focused WP4 source audit, and the remaining sixth on WP5 plus reusable certification. This is not a schedule; it is a hedge against correlated failure. WP1 is the most direct theorem, WP2–WP3 are the most problem-specific arithmetic experiments, and WP4–WP5 search for genuinely new sources of global size.

17 Unified lead inventory and triage

The table records all substantive clusters from the four supplied reports, including calibrated negatives. “Now” means the first experiment can be run using present objects; “conditional” means an explicitly named bridge must first be built; “context” means the result is useful for orientation or for proving a no-go statement.

Cluster	Representative references	Source	Class	Triage / missing bridge
Entire functions sharing integrality arguments	Pila I–II [61, 62]	R4,R3	D	Now. Compute the canonical order-two type and classify forced algebraic dependence.
Concordant/minimal envelopes	Pila–Rodríguez Villegas; Pila; Gramain [60, 58, 38]	R4,R3	A	Now. Verify concordance and obtain a sharp lower-growth benchmark.
Products of two logarithms	Calegari–Dimitrov–Tang [16]	R3,R4	T	Now, focused audit. The theorem excludes the target parameters; seek a uniform arithmetic deformation.
Unbounded denominators / arithmetic holonomy	Calegari–Dimitrov–Tang [17]	R4,R3	T	Conditional. Construct an arithmetic holonomic series attached to the orbit.
Slope/algebraization criteria	Bost; Bost–Charles [11, 12]	R4	T	Conditional. Need an algebraic formal leaf with orbit evaluation maps.
Multivariate fixed divisors	Prasad–Rajkumar–Reddy [63]	R1	A	Now. Compute on S_K with the baseline Newton polytopes.
Total-degree multivariate factorials	Evrard [30]	R1	A	Now. Low-cost first test before the multidegree refinement.
Composite b -orderings	Lagarias–Yangjit [49]	R1	A	Now. Test 6, 12, 18, . . . and weighted ideal families.
Arbitrary-ideal, removed, and order- h orderings	Lagarias–Yangjit [50]	R1	A	Now/conditional. Compute finite invariants; asymptotic gain must be proved.
Adelic regular bases	Chabert–Peruginelli [20]	R1	A	Conditional. Verify residual-cardinality hypotheses for the semigroup closures.
Matroid structure of optimal sets	Grinberg–Petrov [39]	R1	A	Now. Compute common bases and weighted simultaneous defect.
Removed-ordering greedoids	Krachun–Mirgalimova [47]	R1	A	Now. Test robustness under deletion and boundary perturbation.
Algorithms for refined orderings	Johnson [44]	R1	C	Now. Reuse for exact finite experiments.
Simultaneous orderings/equidistribution	Frączyk–Szumowicz; Byszewski et al. [32, 14, 74]	R1	T	Conditional. Derive an energy limit for the semi-group truncations.
Transfinite diameter and resultants	DeMarco–Rumely [26]	R1	T	Conditional. Express the determinant dynamics as a polynomial-map capacity problem.
Adelic Fekete–Szegő theory	Rumely [67]	R1	X	Context. Determine whether local constraints are subcritical, critical, or supercritical in capacity.
Real analytic determinant method	Bombieri–Pila; Pila postulation [10, 59]	R4	T	Now as design. Optimize staircase monomials; no direct curve theorem applies.
p -adic/global determinant method	Heath-Brown; Salberger; Walsh [41, 69, 78]	R3,R4	T	Now as prototype. Force global rank drop before taking archimedean size.
Uniform global-field determinant bounds	Binyamini–Cluckers–Kato; CCDN [8, 15]	R3	T	Conditional. Useful after algebraization of finite orbit blocks.
Alternants for four exponentials	Habsieger [40]	R3	X	Now as audit. Translate the current determinant and isolate the extra hypothesis still missing.
Modular interpolation determinants	Nesterenko [55]	R4	T	Context. Demonstrates the kind of extra differential/modular relation that can create decisive divisibility.
House bounds	Dimitrov; Pritsker [29, 64]	R4	A	Conditional. Need low effective degree and uniform conjugate control.
Parametric geometry of numbers	Schmidt–Summerer; Roy; DFSU [71, 65, 22]	R4	T	Conditional. Translate integrality into a forbidden minima trajectory.
Hypergeometric PGN / shifted logs	Kawashima–Poëls; Hirata-Kohno et al. [46, 42]	R4	T	Conditional. Build a Padé table containing M , A arithmetically.
S -unit Padé approximations	Hirata-Kohno et al. [68]	R4	T	Context/conditional. Finiteness is insufficient; the binomial Padé construction may transfer.
Hankel–Padé irrationality criteria	Bugeaud–Han–Wen–Yao [13]	R3	A	Now as audit. Pair nonvanishing with exact denominator and energy asymptotics.

Cluster	Representative references	Source	Class	Triage / missing bridge
Equilibrium rational approximation	Gonchar–Rakhmanov; Stahl–Totik; Beukers [37, 75, 6]	R4	T	Now as benchmark. Determine whether the current residual is potential-theoretically optimal.
Arithmetic unlikely intersections	Campagna–Dill–Wilms [18]	R4,R1	A	Conditional. First force an algebraic auxiliary curve; dimension two is also only partial.
Effective Pfaffian counting	Binyamini–Jones–Schmidt–Thomas [9]	R4	T	Context. Effectivity does not see denominator support.
Analytic gcd bounds	Pasten–Wang [57]	R4	A	Conditional. Construct two algebraically independent meromorphic functions with a forced excessive gcd counting function.
Integer-valued definable functions	Jones–Thomas–Wilkie; Bhardwaj et al. [45, 7]	R4	A	Conditional. Need a definable integer-valued representative on all $\mathbb{N}$, not just sparse candidates.
Algebraic unlikely intersections	BMZ; Maurin [5, 54]	R2,R3	X	Retire as direct route. The orbit arc is not algebraic.
Complex powers	Gallinaro; Gallinaro–Kirby; Aslanyan–Kirby–Mantova [33, 34, 4]	R4	T	Context. Existence/tameness results do not give arithmetic rigidity.
Drinfeld-module logarithms	Papanikolas; Chang–Papanikolas; Gezmiş–NamoiJam [56, 21, 36]	R3	T	Context. Search for a characteristic-zero substitute for Frobenius difference structure.
Mahler relations and automatic values	Adamczewski–Favrejon [1, 2, 3]	R3	A	Conditional. First derive an arithmetic Mahler equation or simultaneous automaticity.
Digit and carry dynamics	Lagarias; Saye; Dimitrov–Howe [48, 70, 28]	R3	C	Now for certificates. Infinite use needs a synchronized finite-state invariant.
Rank matrices of logarithms	Dasgupta; Dasgupta–Kakde [25, 24]	R2,R3,R4	X	Context. Precisely identifies the balanced 2×2 frontier.
Analytic subgroup / period theorems	Wüstholz; Gaudron–Rémond [79, 35]	R2,R4	X	Conditional. Need an algebraic subgroup arising from integrality, not from a transcendental direction.
Six-exponentials/algebraic-independence ceilings	Roy; Diaz [66, 27]	R2,R3	X	Context. Becomes active only after a third direction or algebraic exponent is forced.
Explicit log 2, log 3 approximation	Simons–de Weger; Simons [73, 72]	R3	C	Now. Improve uniform finite certificates, not the bilinear wall.
Hypergeometric logarithm measures	Zudilin; Marcovecchio [80, 51, 52]	R3	C	Now for Priority H. Verify every joint-pair constant from primary sources.
Formal certificate interoperability	DSLean [23]	R3	C	Now. Translate Gappa/interval certificates to Lean proof terms.
Colossally abundant application	Caveney–Nicolas–Sondow [19]	R3	C	Downstream. Formalize after core definitions stabilize.
Semiabelian gap principle	Hultberg [43]	R1	T	Context. Requires an algebraic subvariety absent from the present arc.
Steering-word complexity	Stephan [76]	R1	X	Context. Needs an exact low-complexity word forced by the counterexample.
Transcendence measures for e^α	Fischler–Rivoal [31]	R1	X	Retire for the wall. α is algebraic in the theorem; the target coefficients are logarithmic.
Massold’s exponential-array estimates	Massold [53]	R1	X	Already covered by baseline. Current parameter range does not settle the balanced 2×2 case.

17.1 Pursue, hold, and retire

Bucket	Contents
Pursue now	Pila growth audit; multivariate factorials; composite/ideal orderings; matroid defect; global p -adic determinant prototype; CDT source audit; exact potential- theory benchmark; finite certificate infrastructure.
Hold for a bridge	House/capacity methods until conjugates are controlled; unlikely intersections until an algebraic curve is forced; Mahler until a functional equation is derived; o-minimal integer-valued theory until a definable representative is constructed; analytic gcd until appropriate functions are built.
Use as context/no-go	Generic BMZ/Zilber–Pink; direct analytic subgroup theorem; ordinary Baker constant improvements; symbolic chaos without automaticity; complex-power existence as a source of arithmetic rigidity; general capacity calculations that remain critical.

18 Conclusions

The four surveys do not uncover a hidden proof, but their critical merge produces a sharper research landscape than any report separately.

First, Pila’s integer-valued-entire-function program is not merely analogous to the present interpolation work: it is formulated on the same rank-two additive semigroup and supplies an explicit order-two threshold. This should be measured immediately.

Second, the generalized-factorial cluster materially extends the ordinary Bhargava analysis already performed. Multivariate fixed divisors, composite ideals, adelic regular bases, and matroidal simultaneous defects are distinct invariants. Their merit is that they can be falsified on finite grids at the correct asymptotic scale.

Third, the CDT product-of-logarithms theorem demonstrates that the terminal type of quantity is not universally inaccessible. What makes its family tractable is arithmetic holonomy, not a better generic Baker constant. The right question is whether the candidate semigroup supports any comparable arithmetic functional equation.

Fourth, global determinant and capacity theories clarify what the present methods are missing. A proof must turn local congruence into a global relation before separate optimization, or control enough conjugates that one exponentially small embedding has global weight. If the corresponding capacity is exactly critical, that fact should be proved and the route closed.

Finally, the negative results are actionable. They rule out spending effort on generic rational-point counting, algebraic unlikely intersections on the transcendental arc, model-theoretic existence theorems, or linear-logarithm refinements that never engage Equation (1). The surviving portfolio is smaller, more arithmetic, and equipped with explicit decision gates.

A Bibliography notes

The bibliography below favors primary papers and stable arXiv records. Several entries are included as methodological templates rather than applicable theorems; their status is stated in the body. Bibliographic corrections described in [Section 3](#) have been incorporated.

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